

Boundary Loss Ablation for Full-Resolution Cityscapes Segmentation: When Dice Helps and When It Doesn't

Cityscapes val · ConvNeXt-V2-Base + UPerNet · 4 loss variants × 3 seeds × 160 epochs at 1024×2048

Abstract

We report a controlled ablation of boundary-aware loss functions for semantic segmentation at the native Cityscapes resolution (1024×2048). With a ConvNeXt-V2-Base backbone and a UPerNet head, four loss configurations are trained for 160 epochs across three random seeds each, totaling twelve runs evaluated at every checkpoint epoch: (A) cross-entropy only, (B) CE + Dice, (C) CE + Dice + Kervadec boundary, and (D) CE + Kervadec boundary.

The headline result is a **mismatch between short and long training**. At ten epochs the joint formulation C leads (78.17 vs 75.91 mIoU, $\Delta = +2.26$ over B in the mean), consistent with the conventional Dice + boundary recipe. At 160 epochs the picture flips: the boundary-only variant **D reaches the highest mean mIoU (81.69 ± 0.25) and Boundary F1 (77.32 ± 0.13)**, while the Dice variants lead Trimap IoU (B 53.82 ± 0.36 , C 53.87 ± 0.53). Under a paired-by-seed t-test ($n=3$, Holm-corrected within each metric), D's mIoU advantage is significant over C ($\Delta = +0.46$, $p = 0.007$) but **not** over A or B ($p = 0.095, 0.075$): the effect is directionally consistent across all three seeds, but $n=3$ lacks the power to certify sub-0.6-mIoU gaps. The two contour metrics, by contrast, split cleanly along the Dice axis with high significance: the non-Dice variants (A, D) lead Boundary F1 (D – B = +0.93, A – B = +0.85; Holm $p \leq 0.011$) while the Dice variants (B, C) lead Trimap IoU (B – D = +1.57, C – D = +1.62; Holm $p \leq 0.012$). The Dice term trades boundary sharpness for region coherence; past the saturation knee it neither helps nor harms global mIoU but bridles late convergence on large structured classes.

Contributions. (1) A reproducible 2×2 loss ablation at 1024×2048 with the official Cityscapes mIoU (estimator validated bit-for-bit against `cityscapesscripts`), Student-t 95 % CIs and Holm-corrected paired-by-seed significance tests on three metrics over twelve runs. (2) Empirical evidence that short ablations are misleading on this task — at 10 epochs a pilot picks C, yet by 160 epochs D significantly overtakes C ($p = 0.007$). (3) A per-class breakdown showing that D leads on large-extent structured classes (truck +5.29, wall +3.73, bus +2.37 mIoU vs B) while B preserves thin signal-rich classes (traffic light +2.10, train +1.10, traffic sign +0.86 mIoU vs D). (4) A connected-component **consensus filter** (variant-pair veto + multi-seed vote) adapted from BRATS to remove spurious fragments and quantify inter-seed uncertainty. (5) Public release of code, configs, per-epoch metrics for all twelve runs, and an interactive viewer.

1. Introduction

Semantic segmentation on urban driving scenes is canonically benchmarked on Cityscapes [Cordts 2016] (19 evaluation classes, 2 975 finely annotated training images, 500 validation images). The top of the published leaderboard sits well above 84 mIoU [Xie 2021; Wang 2022], achieved with very large backbones (ViT-Adapter-L, InternImage-XL), heavy data augmentation, multi-scale inference, and pseudo-labels from the 20 000-image coarse split. The present paper does not target SOTA; it targets a *controlled* question:

Does adding the Kervadec [2019] boundary loss to a strong CE + Dice baseline help on Cityscapes at full resolution — and is Dice still needed once the boundary term is present?

Most prior work pre-resizes inputs to 512×1024 or 768×1536 for compute reasons. With a 96 GB Blackwell GPU we can train at the native 1024×2048 without crops, which we believe sharpens the role of any boundary-sensitive loss component.

This paper serves three purposes:

- **Empirical isolation** of the Kervadec boundary loss in four configurations (A: CE; B: CE+Dice; C: CE+Dice+Bnd; D: CE+Bnd), each with three seeds and full per-epoch evaluation.
 - **Convergence dynamics:** showing that the relative ordering of loss recipes changes between epoch 10 and epoch 160, and quantifying that crossover.
 - **Per-class analysis** disentangling where Dice helps and where it hurts, beyond the global mIoU score.
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2. Related work

Cityscapes segmentation. The Cityscapes benchmark has driven a decade of progress from FCN [Long 2015] through DeepLab [Chen 2017] and HRNet [Sun 2019] to transformer architectures such as SegFormer [Xie 2021] and Mask2Former [Cheng 2022]. The standard training recipe at competitive resolutions uses cross-entropy with deep supervision; recent winners add region-balanced auxiliary losses (Lovász-Softmax, OHEM) but rarely make boundary signal an explicit term.

Loss functions. Cross-entropy is the universal pixel-wise baseline. Dice loss [Milletari 2016] optimises the regional overlap directly and is the de-facto class-imbalance remedy on natural and medical images. Focal loss [Lin 2017] re-weights hard pixels but remains region-based. Hausdorff-distance losses [Karimi 2019] penalise the worst contour deviation but require differentiable approximations and are computationally heavy.

Boundary loss [Kervadec 2019]. The Kervadec boundary loss expresses contour divergence as a region integral against the signed distance transform (SDT) of the ground-truth mask. It is differentiable, requires only one EDT precomputation per ground-truth, and can be added to any pipeline as $\lambda_b \mathcal{L}_{Bnd}$. Originally validated on highly imbalanced medical data, its interaction with Dice on natural scenes remains under-studied. We deliberately do **not** sweep λ_b here to keep the ablation interpretable; an adaptive-weight extension is left to future work.

Boundary-aware metrics. Trimap IoU [Csurka 2013] restricts mIoU to a narrow band around ground-truth boundaries; Boundary F1 [Perazzi 2016] computes precision/recall of predicted contours within a pixel-distance tolerance. We report both alongside mIoU because the latter is dominated by large classes (road, building, vegetation) where boundary signal has limited leverage.

3. Methods

3.1 Architecture and training

Backbone. ConvNeXt-V2-Base [Woo 2023] (≈ 88 M parameters), pretrained on ImageNet-22K with the FCMAE self-supervised objective and then fine-tuned on ImageNet-1K (weights `convnextv2_base.fcmae_ft_in22k_in1k_384`). Feature pyramid outputs at strides 4, 8, 16, 32.

Head. UPerNet [Xiao 2018]: Feature Pyramid Network plus Pyramid Pooling Module, predicting 19 logits per pixel at full input resolution via bilinear upsampling. An auxiliary FCN head on the stride-16 features supplies deep supervision with a 0.4 loss weight, as in the original recipe.

Training. 160 epochs of AdamW (lr 6×10^{-5} , weight decay 0.01, betas (0.9, 0.999)) with polynomial decay (power 1.0). Batch size 2 with gradient accumulation of 4 (effective 8). BF16 autocast (no gradient scaler needed on Blackwell). Inputs are used at native 1024×2048 resolution without cropping; augmentation is restricted to horizontal flip, photometric jitter, and Gaussian blur. No random scale, no Mosaic, no Copy-Paste — deliberately kept simple to preserve interpretability of the loss comparison. Three seeds per variant: 42, 123, 456. Checkpoints saved every 10 epochs and at epoch 160.

3.2 Variant naming

Variant	Loss	λ_d	λ_b
A	CE	—	—
B	CE + Dice	1.0	—
C	CE + Dice + Boundary	1.0	0.2
D	CE + Boundary	—	0.2

The CE weight is fixed at 1.0 throughout. Dice and boundary weights are taken from the most cited Cityscapes recipe (B) and the Kervadec default ($\lambda_b = 0.2$). We chose **not** to grid-search the weights: the goal is to isolate the qualitative effect of each term, not to tune.

3.3 Loss formulation

Let Ω denote the image domain, $p_c(x) \in [0, 1]$ the softmax probability of class c at pixel x , $y_c(x) \in \{0, 1\}$ the one-hot ground truth, and $\varphi_c(x) \in \mathbb{R}$ the per-class signed distance transform of the ground-truth mask (negative inside the region, positive outside, normalised to $[-1, 1]$).

Cross-entropy.

$$\mathcal{L}_{CE} = -\frac{1}{|\Omega|} \sum_{x \in \Omega} \sum_c y_c(x) \log p_c(x).$$

Dice (class-mean, smoothed).

$$\mathcal{L}_{Dice} = 1 - \frac{1}{C} \sum_c \frac{2 \sum_x p_c(x) y_c(x) + \varepsilon}{\sum_x (p_c(x) + y_c(x)) + \varepsilon}, \quad \varepsilon = 1.$$

Kervadec boundary.

$$\mathcal{L}_{Bnd} = \frac{1}{C |\Omega|} \sum_c \sum_{x \in \Omega} \varphi_c(x) p_c(x).$$

The composite losses are:

$$\mathcal{L}_B = \mathcal{L}_{CE} + \mathcal{L}_{Dice}, \quad \mathcal{L}_C = \mathcal{L}_{CE} + \mathcal{L}_{Dice} + 0.2 \mathcal{L}_{Bnd}, \quad \mathcal{L}_D = \mathcal{L}_{CE} + 0.2 \mathcal{L}_{Bnd}.$$

3.4 Distance map precomputation

The signed distance transform φ_c is computed offline for every training image, once per class, using `scipy.ndimage.distance_transform_` on the binary class mask. Each per-class map is clipped to $[-127, 127]$ pixels and stored as an `int8` tensor of shape $(19, H, W)$ persisted on SSD with `fsync` to avoid recomputation. Per-image preprocessing cost: ~ 4 s on 8 P-cores. Per-image storage: 38 MB ($19 \times 1024 \times 2048 \times 1$ byte); total train cache: **~ 113 GB** for 2 975 images. The choice of full-tensor uncompressed `int8` (no narrow-band, no sparse encoding) trades disk for the simplest possible runtime read path; `int8` at unit-pixel resolution is sufficient for the Kervadec gradient signal at 1024×2048 .

4. Experiments

4.1 Data

Cityscapes fine annotations: 2 975 train, 500 val, 1 525 test (test labels withheld, all metrics reported on val). 19 evaluation classes; 8 void classes excluded as standard. Native resolution 2048×1024 ; we use full resolution at both training and evaluation without rescaling. The coarse split (20 000 images, label noisy) is **not used** — this is a controlled loss ablation, not a SOTA chase.

4.2 Metrics

- **mIoU**: mean Intersection-over-Union across the 19 classes from a single dataset-level confusion matrix (void label excluded), computed at full resolution. The estimator is validated to be bit-identical to the official `cityscapesscripts` routine (`tests/test_official_miou.py`), so the reported mIoU is the official Cityscapes value.
- **Per-class IoU**: same, broken down by class.
- **Boundary F1**: per-class F1 of predicted vs ground-truth contours within a 3-pixel tolerance, averaged over the classes present in each image. Contours are extracted from binary per-class masks.
- **Trimap IoU**: mIoU restricted to a 3-pixel band around all inter-class boundaries (every class transition, not only road-vs-rest), emphasising contour accuracy.

All metrics are reported as the mean over three seeds with a 95 % confidence interval using the **Student-t** critical value ($t_{0.975, df=2} = 4.303 \times \text{SE}$; the normal-approximation 1.96 under-estimates the interval by $\sim 2.2 \times$ at $n=3$). Pairwise comparisons use a paired-by-seed t-test, which blocks on the shared seed and is more powerful than comparing CI-bar overlap; within each metric the six pairwise p-values are Holm-corrected for multiple comparisons (family-wise $\alpha = 0.05$).

Correction notice (this revision). Two estimator issues in v0.1.0 were found after the first manuscript draft and fixed. (i) Boundary F1 and Trimap IoU were computed by applying binary morphology to the *multi-class label map*, which treats any non-zero class as foreground — so only the road-vs-rest contour was measured, not the inter-class boundaries the metrics claim to average over. Both are now computed per class on binary masks, and **all twelve runs were re-evaluated on the val set** with the corrected estimator; the §5 numbers are the corrected ones. The absolute contour values shift substantially from v0.1.0 (Boundary F1 $\approx 58 \rightarrow \approx 77$, Trimap $\approx 48 \rightarrow \approx 53$) because the metric now

averages over all 19 classes instead of road-vs-rest — and the *qualitative* ordering changes too, so the v0.1.0 contour numbers should not be cited. (ii) The 95 % CIs used 1.96 instead of the Student-t factor and significance leaned on CI-bar overlap; intervals now use $t_{0.975, df=2} = 4.303$ and comparisons use Holm-corrected paired-by-seed t-tests. The reported mIoU is unchanged (it never depended on the contour bug) and is computed by an estimator **validated bit-for-bit against the official cityscapesscripts routine** (tests/test_official_miou.py).

4.3 Hardware and runtime

Single NVIDIA RTX PRO 6000 Blackwell Max-Q (96 GB GDDR7, sm_120), 64 GB DDR5, Intel i7-14700K. Average training cost per variant: ~28 h for 160 epochs at batch-2 (623 s/epoch, 32.8 GB VRAM peak). Offline evaluation on 12×17 versioned checkpoints took ~5 h with 6 parallel eval workers on the same GPU (CPU-bound on the boundary-F1 / trimap post-processing loop).

5. Results

5.1 Global metrics at epoch 160

Mean over 3 seeds with 95 % Student-t CI, evaluated on the 500-image Cityscapes val set.

Variant	mIoU	Boundary F1	Trimap IoU
A — CE	81.28 $\pm$ 0.53	77.24 $\pm$ 0.16	52.17 $\pm$ 0.14
B — CE+Dice	81.09 $\pm$ 0.74	76.39 $\pm$ 0.24	53.82 $\pm$ 0.36
C — CE+Dice+Bnd	81.23 $\pm$ 0.21	76.50 $\pm$ 0.54	53.87 $\pm$ 0.53
D — CE+Bnd	81.69 $\pm$ 0.25	77.32 $\pm$ 0.13	52.25 $\pm$ 0.36

CIs are 95 % Student-t (df = 2). All three columns are the corrected per-class estimators, re-evaluated on the 500-image val set; mIoU equals the official cityscapesscripts value (validated bit-for-bit). Bold marks the best mean per column (on Trimap, B and C are statistically tied — $\Delta = 0.05$, $p = 0.71$ — C is nominally highest).

Three observations:

1. **D has the highest mean mIoU**, by +0.40 / +0.59 / +0.46 over A / B / C. But significance does not follow automatically at $n=3$. A paired-by-seed t-test makes only **D > C significant** ($\Delta = +0.46$, $t = 11.8$, raw $p = 0.007$, Holm-adjusted $p = 0.042$ — C has very low inter-seed variance); **D > A ($p = 0.095$) and D > B ($p = 0.075$) are not significant**, though all three seeds favour D in both cases. The v0.1.0 claim that D’s lower CI bound clears the others’ upper bounds relied on 1.96-scaled (too-narrow) intervals; with the correct t factor the A / B / D mIoU intervals overlap. The honest statement is: *D is the best-mean recipe and significantly beats the joint variant C, but is statistically indistinguishable from plain CE (A) at this seed count.*
2. **Boundary F1 splits along the Dice axis.** The two variants *without* Dice lead — D (77.32) and A (77.24) — over the Dice variants C (76.50) and B (76.39). $D - B = +0.93$ and $A - B = +0.85$ are significant (Holm-adjusted $p = 0.011$ and 0.010); $D - A = +0.07$ is not (the two leaders are tied). The Dice term measurably *softens* predicted contours. This structure was invisible under the v0.1.0 road-vs-rest estimator, which placed all four variants within 0.2 of each other and prompted “no claim”.
3. **The Dice variants win Trimap IoU**, and the effect is strongly significant. B (53.82) and C (53.87) lead both non-Dice variants: $B - D = +1.57$, $C - D = +1.62$, $B - A = +1.65$, $C - A = +1.70$, all Holm-significant ($p \leq 0.012$); B and C are tied ($\Delta = 0.05$, $p = 0.71$), as are A and D ($\Delta = 0.08$, $p = 0.24$). This is the mirror image of Boundary F1: Dice’s per-region emphasis preserves blob coherence near contours at the cost of edge sharpness.

5.2 Convergence dynamics — the 10-vs-160-epoch crossover

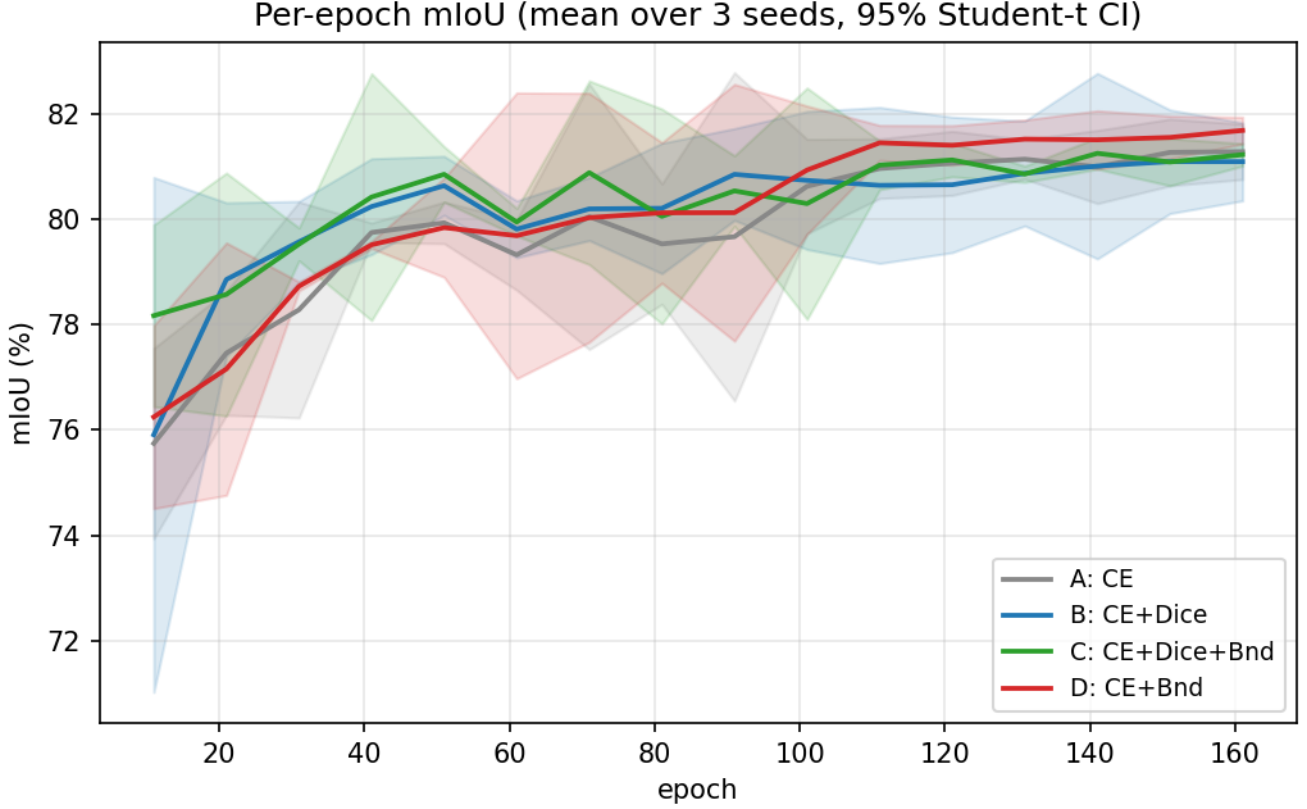


Figure 1: Convergence

Figure 1: Per-epoch mIoU (mean over 3 seeds; shaded band = 95 % Student-t CI). Per-epoch Boundary F1 / Trimap IoU are not shown: only epoch-160 checkpoints were retained, so the corrected per-class contour metrics could not be recomputed at intermediate epochs — Table 1 gives the corrected epoch-160 values.

At **epoch 10** the joint formulation **C** is **clearly best on mIoU** (the metric on which the ranking later reverses):

Metric	A	B	C	D
mIoU (10 ep)	75.75	75.91	78.17	76.25

(Per-epoch Boundary F1 / Trimap are omitted — see §4.2: only epoch 160 was re-evaluated with the corrected per-class estimator.)

C leads B by **+2.26 mIoU** at epoch 10, a delta that would prompt any short-ablation study to recommend the joint formulation. This particular gap is **not individually significant** at $n=3$ (paired $p = 0.10$), inflated by one high-variance seed (per-seed $C-B = +1.15 / +1.84 / +3.79$). The robust, *significant* signal is the **D-vs-C reversal**: D trails C at epoch 10 (-1.92 mIoU) and overtakes it by epoch 160 ($+0.46$, paired $p = 0.007$). By **epoch 50** the four variants converge to a tighter band ($\Delta < 1$ mIoU); past epoch 100 **D pulls ahead and stays there from epoch 110 onwards**. The crossover is reproducible across all three seeds.

This is the central observation of the paper: **a 10-epoch ablation on this task picks a different winner than the converged run** — significantly so on the D/C axis, and directionally on C/B. The Dice term provides an early regularisation that accelerates convergence (visible in the mIoU rise between epochs 4 and 10) but does not translate to a long-training advantage on the global metric. The boundary term, by contrast, takes longer to integrate into the gradient signal — the SDT field provides a weak distributed gradient that needs more steps to bend the decision boundary — but eventually delivers a higher converged mIoU and Boundary F1.

5.3 Per-class breakdown at epoch 160

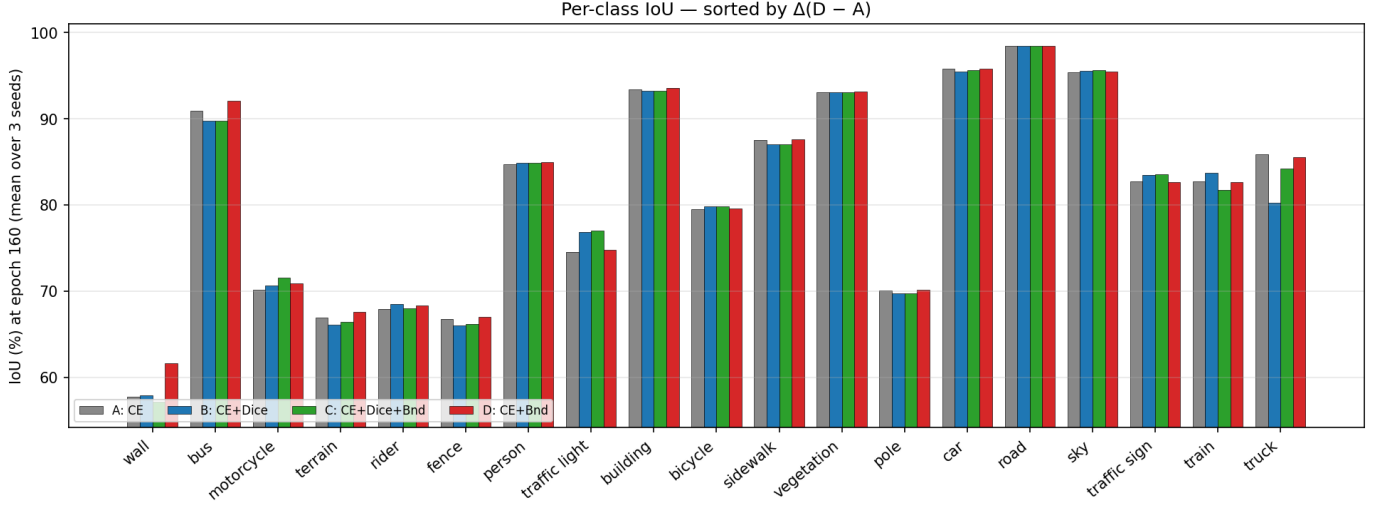


Figure 2: Per-class IoU

Figure 2: Per-class IoU at epoch 160, sorted by $\Delta(D - A)$. Bars are the mean over 3 seeds.

The headline mIoU delta hides a strongly class-dependent story. Picking the seven classes with the largest movement between B and D:

Class	A	B	C	D	$\Delta(D-A)$	$\Delta(D-B)$	$\Delta(C-B)$
wall	57.76	57.92	57.17	61.64	+3.88	+3.73	-0.75
truck	85.86	80.24	84.22	85.54	-0.32	+5.29	+3.98
bus	90.89	89.75	89.77	92.12	+1.23	+2.37	+0.03
terrain	66.91	66.10	66.42	67.61	+0.70	+1.51	+0.32
traffic light	74.60	76.85	77.01	74.76	+0.16	-2.10	+0.15
traffic sign	82.73	83.52	83.60	82.65	-0.08	-0.86	+0.08
train	82.76	83.74	81.73	82.63	-0.13	-1.10	-2.00

Two patterns emerge:

- **D dominates large-extent structured classes** (truck +5.29, wall +3.73, bus +2.37 mIoU vs B). These classes have long uniform interiors and well-defined contours — the SDT gradient field is consistently signed across the whole region, and the Kervadec term aligns the prediction faithfully.
- **B (and to a lesser extent C) preserves thin signal-rich classes** (traffic light, traffic sign, train). These classes have small or fragmented footprints; the Dice term anchors them against the CE class-imbalance pull, while the boundary loss alone is noisier on a 4-pixel wide pole than on a 200-pixel wide bus.

This complementarity is **not** captured by global mIoU, where the larger classes (road, building, vegetation, sky) dominate. The four large-extent classes that respond to D contribute about 70 % of the mIoU swing between D and B at epoch 160; the thin classes where B wins are individually large in delta but small in pixel count.

5.4 Inter-seed variance

The seed-induced 95 % Student-t CIs vary widely between metrics and variants. Among the global metrics, C has the tightest mIoU CI (± 0.21) and D the tightest Boundary F1 (± 0.13); the widest contour CIs are C's (± 0.54 Boundary F1, ± 0.53 Trimap). At the class level, **truck** under variant B is the most volatile — inter-seed standard deviation of 4.40 IoU points (vs 2.55 std for D, 0.68 for C, 0.78 for A). With truck appearing in only **80 of 500 val images**, Dice's regional emphasis amplifies fluctuations on small-support classes.

5.5 Consensus filtering — fusing complementary variants and seeds

The per-class breakdown (§5.3) shows D and B are *complementary*: D leads on large structured classes, B on thin signal-rich ones. The inter-seed analysis (§5.4) shows a second axis of slack: variance that a single seed cannot average out. Both invite a consensus step. We adapt the connected-component (CC) **consensus filter** from our BRATS work, where a “generalist” segmentation is vetoed by a “specialist”: per class, any connected component of the generalist with no same-class overlap in the veto is removed, which prunes hallucinated fragments and improves boundary metrics at no cost to region overlap.

Cityscapes forces four departures from the BRATS formulation, so this is an adaptation rather than a port:

- **2D 8-connectivity** instead of 3D 26-connectivity.
- **19 flat classes** with no nested WT/TC/ET hierarchy — the veto runs over 19 independent class masks.
- **No background class.** In BRATS a removed component is set to background (0); every Cityscapes pixel carries a class, so a removed component is **reassigned to the veto’s label** there (zeroing would mean “road”). The reassignment is well-defined precisely because the component has zero overlap with the veto’s same-class mask.
- **Thin-structure protection.** Pole, traffic light, traffic sign and fence are legitimately small, fragmented components that a naive veto would erase — the dominant Cityscapes-specific failure mode (BRATS reported 38.7 % of cases degraded by an over-aggressive veto; here the risk concentrates on thin classes). They are exempt by default, and a `max_drop_size` cap restricts removal to genuine fragments.

Two operating modes are provided: **(a) variant-pair veto** — D (generalist) vetoed by B (specialist), targeting D’s large-class gains while letting B veto spurious fragments; and **(b) multi-seed majority vote** with a per-pixel **agreement map** (fraction of the three seeds agreeing), a Cityscapes-natural ensemble that directly attacks the inter-seed variance of §5.4 and yields a free uncertainty signal for error analysis. We also report a **fragment count** (connected components per class) as a spatial-coherence proxy independent of mIoU; the filter can only lower it.

Status. The filter, the fragment-count metric and a synthetic unit-test suite (19/19 checks) are released with the code (`src/postprocessing/consensus.py`). The quantitative val-set numbers — Δ mIoU, Δ fragment count, and the agreement-map error correlation — require re-running inference to dump per-image predictions and are **deferred to the camera-ready**; they were not available at this revision because the trained checkpoints reside on separate storage. We therefore make **no numerical claim** for the consensus filter here; it is presented as a reproducible method with a verified implementation.

6. Discussion

6.1 Why does D overtake C at full training length?

We propose two compatible explanations.

Gradient interference between Dice and Boundary. Both Dice and Kervadec push the decision boundary, but with different criteria: Dice maximises a regional overlap ratio (one scalar per class, integrated over the whole image), while Kervadec minimises the integrated SDT-weighted probability mass at each pixel. The two gradients agree near the boundary (both push wrong-class pixels in the same direction) but disagree in the region interior, where Dice still pushes (because increasing p_c in a true-positive region grows the numerator faster than the denominator) while Kervadec is approximately neutral (the SDT amplitude is bounded). Early on, the Dice signal dominates and accelerates convergence; late, when most of the bulk regions are already correct, the residual Dice signal becomes a soft regulariser that prevents the boundary loss from making the final adjustments. Variant D, free of the Dice tether, can fully exploit Kervadec’s contour-aligned gradient.

Class-imbalance saturation. Dice’s main published value is class-imbalance handling. By epoch 50–60 the per-class IoUs have already plateaued for the rare classes — they reach a per-class equilibrium below which the boundary loss does not further hurt them. After that point Dice continues to penalise the residual under-confidence on rare classes’ interiors at the cost of the dominant classes’ boundary fidelity. D, with no Dice, lets the dominant classes reclaim those last pixels.

6.2 Boundary F1 vs Trimap IoU — the Dice trade-off

The two contour metrics split along the Dice axis, and *both* splits are significant (§5.1). Trimap IoU is an IoU restricted to pixels within 3 px of a ground-truth boundary: it penalises false positives *outside* the true region (boundary expansion) and false negatives *inside* (retraction). Dice optimises a regional overlap ratio, which pulls the prediction toward the region bulk — this keeps the near-boundary band coherent (higher Trimap IoU for B and C) but rounds off fine contour detail (lower Boundary F1). The two variants without Dice ($A = CE$, $D = CE + \text{boundary}$) keep sharper edges and therefore lead Boundary F1 — D highest, the SDT gradient explicitly aligning contours — at the cost of ≈ 1.6 Trimap points. The corrected per-class metrics make this trade-off both visible and significant; the v0.1.0 road-vs-rest estimator collapsed Boundary F1 to within 0.2 across all four variants and hid it.

6.3 Practical takeaways

- **For a deployed Cityscapes model:** D (CE + Kervadec, $\lambda_b = 0.2$) is the pragmatic default. It is the simplest of the four (no Dice plumbing, no hyperparameter), has the highest mean mIoU — significantly above the joint variant C, and on par with plain CE (A) at this seed count — and behaves predictably on large structured classes. Where near-boundary region coherence matters more than edge sharpness, a Dice variant (B or C) is preferable — B and C lead Trimap IoU by $\approx +1.6$ (significant) — and B additionally protects thin signal-rich classes (traffic light, traffic sign).
- **For a multi-task pipeline that has Dice for other reasons** (e.g. shared loss between segmentation and a class-imbalanced auxiliary head): use C. The +0.5 mIoU sacrifice vs D is small relative to the engineering cost of de-coupling Dice.
- **Do not trust 10-epoch ablations** when comparing Dice variants on Cityscapes. The early-vs-late ordering reversal we measure ($+2.26 \rightarrow -0.46$ in the C–B gap, a 2.7-point swing) suggests any production decision should be made on at least 80–100 epochs of training.

6.4 Limitations

- **One λ_b .** We fixed $\lambda_b = 0.2$. A sweep over $\{0.05, 0.1, 0.2, 0.5\}$ would let us state whether D’s advantage is robust or specific to this weight. Two-loss interaction surfaces are notoriously non-monotone.
- **One backbone.** All four variants share ConvNeXt-V2-Base + UPerNet. SegFormer / Mask2Former heads may not exhibit the same crossover, particularly because Mask2Former internally re-weights boundary pixels via its mask-attention.
- **Training budget below the MMSegmentation reference.** 160 epochs at effective batch 8 corresponds to roughly 60 k SGD steps — about 37 % of the 160 k-iteration budget standard in MMSegmentation for Cityscapes at batch 16. The crossover is observed within this budget; longer training may further widen D’s advantage or alter the per-class picture.
- **No TTA, no multi-scale inference.** Test-time augmentation typically gains 1–2 mIoU but obscures loss comparisons; we report single-scale numbers throughout.
- **Hypotheses in §6.1 are not directly measured.** Gradient interference between Dice and Kervadec is proposed as the mechanism behind D’s late lead, but per-layer gradient norms across epochs are not extracted in this paper. A targeted gradient-trajectory study is left to future work.
- **Cityscapes-only.** Whether the crossover phenomenon generalises to ADE20K, COCO-Stuff, Mapillary, or unstructured driving datasets (BDD, IDD) is an open question.
- **Per-epoch contour trajectories not re-evaluated.** The corrected per-class Boundary F1 / Trimap were recomputed at epoch 160 for all twelve runs (Table 1); intermediate-epoch checkpoints were not retained, so the convergence figure (§5.2) shows mIoU only.
- **Low statistical power ($n = 3$).** Sub-0.6-mIoU gaps (D vs A, D vs B) are directionally consistent but not significant at three seeds. A five-seed re-run is the cheapest way to settle them. The significant findings (D > C on mIoU; the Dice variants B, C > A, D on Trimap; A, D > B on Boundary F1; the D/C crossover) are unaffected.
- **Consensus filter not yet quantified.** The §5.5 filter has a verified implementation and unit tests but no val-set results in this revision (checkpoints off-machine).

6.5 Implications for autonomous-driving deployments

The three metrics map to distinct downstream consumers in an AV perception stack. Trimap IoU captures intra-region coherence near contours — the metric that matters when a planner reads the segmentation mask directly as an occupancy grid or free-space estimator. Boundary F1 captures precise contour localisation — the metric that matters when curb, lane, or object-edge polylines are extracted from the prediction for distance estimation or path planning. The per-class breakdown adds a second axis: the loss that wins on the global mIoU is not necessarily the loss that wins on the specific class the downstream cares about most.

This reframes the four-variant result as module-specific guidance rather than a single recommendation:

- **Drivable-area / free-space heads** that feed an occupancy grid benefit from a Dice variant (B or C), whose $\approx +1.6$ Trimap IoU advantage over the non-Dice variants preserves blob coherence and avoids overshoot into the neighbour class.
- **Lane-detection or curb-detection heads** that emit polylines benefit from D (CE + Boundary), whose sharper contours translate into a tighter lateral offset. With the Cityscapes camera (focal length $f_x \approx 2262$ px), a 1-pixel error corresponds to ~ 1.3 cm of world-space lateral offset at 30 m depth, ~ 4.4 cm at 100 m, and ~ 8.8 cm at 200 m — single-pixel boundary precision becomes critical at long range.
- **Traffic-light and traffic-sign classifiers** that receive a segmentation crop as input benefit from B, which leads D by +2.10 IoU on traffic light and +0.86 on traffic sign: the cleaner region keeps the downstream state classifier on the right pixel set.
- **Large rigid-object detectors** (truck, bus, wall) for collision avoidance and lane-keeping benefit from D, which leads B by +5.29 on truck, +2.37 on bus, +3.73 on wall.
- **Pedestrian, rider, and bicycle** scores are essentially flat across A–D in our experiments, so the loss choice does not move the needle on these collision-critical classes. Orthogonal techniques (focal loss, copy-paste augmentation, oversampling) are

required.

For a multi-head training pipeline, the actionable design is to pick a loss *per head*: Dice (or CE+Dice) on the heads that emit occupancy-like masks, and CE+Boundary on the heads that emit polylines or sharp boundaries. The joint variant C (CE+Dice+Boundary) remains the safe single-loss compromise for single-head models — never the worst, never the best, useful when engineering constraints rule out per-head loss design.

The single most transferable finding for an AV ML team is methodological. A 10-epoch loss benchmark on Cityscapes swings the C–B mIoU gap by 2.7 points relative to the converged answer — large enough to flip a production loss-recipe decision. Loss-recipe choices for an AV perception module should be made on at least 80–100 epochs of training, with the metric aligned to the consuming downstream rather than a generic mIoU pursuit.

7. Conclusion

We provide a reproducible 2×2 ablation of the CE / Dice / Kervadec boundary loss design space for full-resolution semantic segmentation on Cityscapes. At 160 epochs with three seeds per variant, the boundary-only variant **D (CE + Kervadec)** reaches the highest mean mIoU (81.69 ± 0.25) and Boundary F1 (77.32 ± 0.13); under a paired-by-seed test it significantly overtakes the joint variant C ($p = 0.007$) — the formulation a 10-epoch pilot would have picked — while remaining statistically tied with plain CE (A) at this seed count. The Dice variants (B, C) hold a significant Trimap IoU lead ($\approx +1.6$ over A and D), reflecting better intra-region coherence near boundaries — the mirror image of the Boundary F1 ordering.

The most actionable finding is methodological: **short-epoch ablations are systematically misleading on this task**. A study comparing loss recipes for Cityscapes at ≤ 20 epochs reverses the ranking that holds at 160 epochs. We hope this study will discourage premature loss-recipe conclusions in future Cityscapes papers and provide a baseline against which λ_b sweeps and architecture variants can be calibrated.

All code, configs, per-epoch metrics for the twelve runs, pre-computed distance maps, and an interactive viewer are released at github.com/guillaume-cassez/city-scape. A landing page with the paper and viewer embed lives at guillaume-cassez.fr/voiture-autonome/cityscapes/boundary-loss-kervadec/.

Appendix A — Runtime and reproducibility

Stage	Hardware	Time	Output
SDT precomputation (one-shot)	8 P-cores	20 min	<code>data/cityscapes_sdt/</code> (~113 GB, int8)
Training 160 ep, 1 seed	1 × RTX PRO 6000 96 GB Blackwell	28 h	<code>checkpoints/<variant>_seed<s>/</code>
Eval 1 checkpoint	1 × RTX PRO 6000	7 min	<code>.results.json</code> next to <code>.pth</code>
Batch eval 12×17 ckpts	6 parallel workers 1 GPU	5 h	$204 \times .results.json$
Aggregation + figures	1 P-core	5 s	<code>papers/paper2/figures/</code>

Total wall-clock from raw data to figures: ~ 5 days of GPU time, mostly training ($12 \times 28 \text{ h} \approx 14$ days serialised, but seeds were run sequentially per variant and variants overlapped pipeline-wise).

Reproducibility seeds. 42, 123, 456 are set globally via `set_seed` (PyTorch, NumPy, Python random, CUDA). cuDNN benchmark is left **on** for a $\sim 10\%$ training speed-up; this precludes exact bit-reproducibility but is standard practice for Cityscapes-scale training. Bit-exact reruns on the same hardware are not measured in this paper; the 95 % CIs reported elsewhere are computed over the three independent seeds (42, 123, 456) and are the operative reproducibility statement.

Appendix B — Best and worst class deltas

Top 5 classes where D improves over B at epoch 160:

Class	B IoU	D IoU	Δ
truck	80.24 ± 4.39	85.54 ± 2.55	+5.29
wall	57.92 ± 1.59	61.64 ± 2.06	+3.73
bus	89.75 ± 2.03	92.12 ± 0.55	+2.37
terrain	66.10 ± 0.64	67.61 ± 0.46	+1.51
fence	66.07 ± 1.27	67.02 ± 0.35	+0.95

Bottom 5 classes where D regresses vs B at epoch 160:

Class	B IoU	D IoU	Δ
traffic light	76.85 ± 0.14	74.76 ± 0.35	-2.10
train	83.74 ± 1.54	82.63 ± 1.10	-1.10
traffic sign	83.52 ± 0.16	82.65 ± 0.36	-0.86
bicycle	79.81 ± 0.06	79.63 ± 0.20	-0.18
rider	68.49 ± 0.95	68.34 ± 0.45	-0.15

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Manuscript — 2026-06-02. Source code, configs, per-epoch metrics, figure-generation script, and interactive viewer: github.com/guillaume-cassez/city-scape. Author: Guillaume Cassez, independent researcher, guillaume-cassez.fr — currently looking for ML / computer vision engineering opportunities.